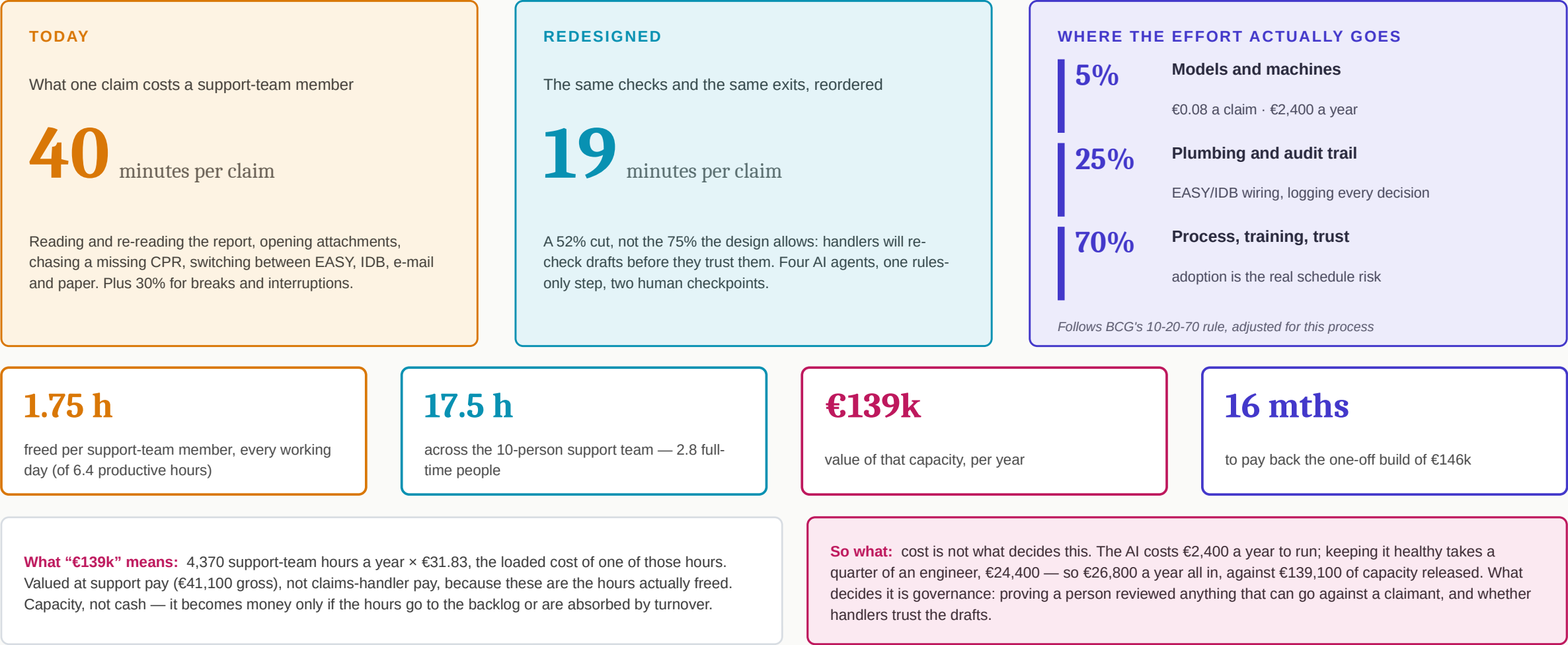


EXECUTIVE SUMMARY

Redesigning intake gives the 10-person support team back 1.75 hours each, every day

17.5 hours a day across the team — 2.8 full-time people · two thirds of that work goes to claims that were always going to close quietly · a person still decides anything adverse



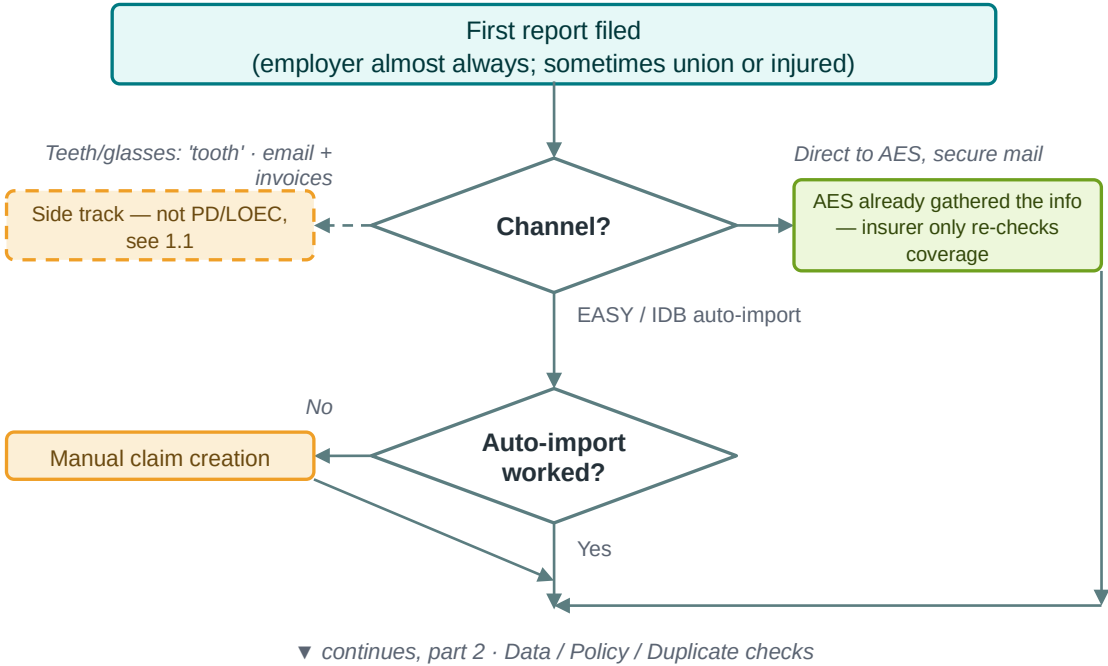
Conservative throughout: the 21-minute-per-claim saving is a three-point estimate (best 30 / likely 21 / worst 12), not the design maximum. Build-up on slide 9, sources on slide 10.

Three ways in and a separate side-track diagram become one classified intake

FNOL as it runs today (left) against the redesign (right) · first report to an opened, policy-linked claim

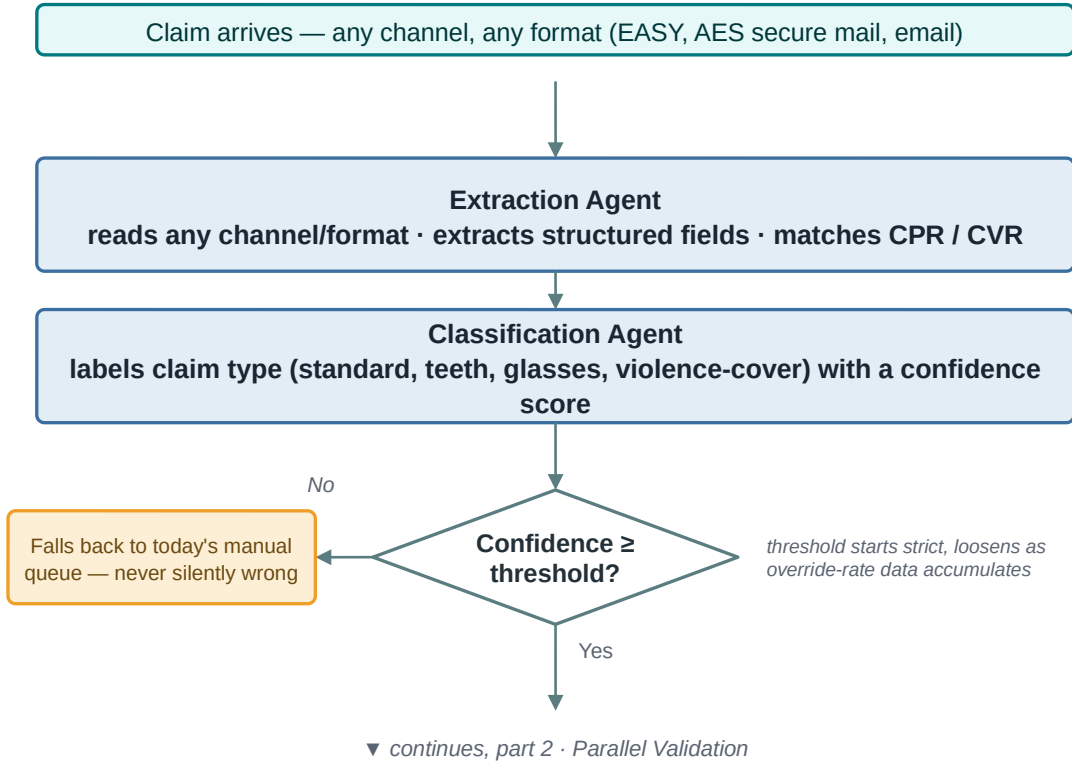
Trigger / start Step / decision Exception / rework Exit

TODAY



AI agent (LLM call) Rules only, no model Human checkpoint

AI-NATIVE — one LangGraph state graph · shared claim state carries the audit trail



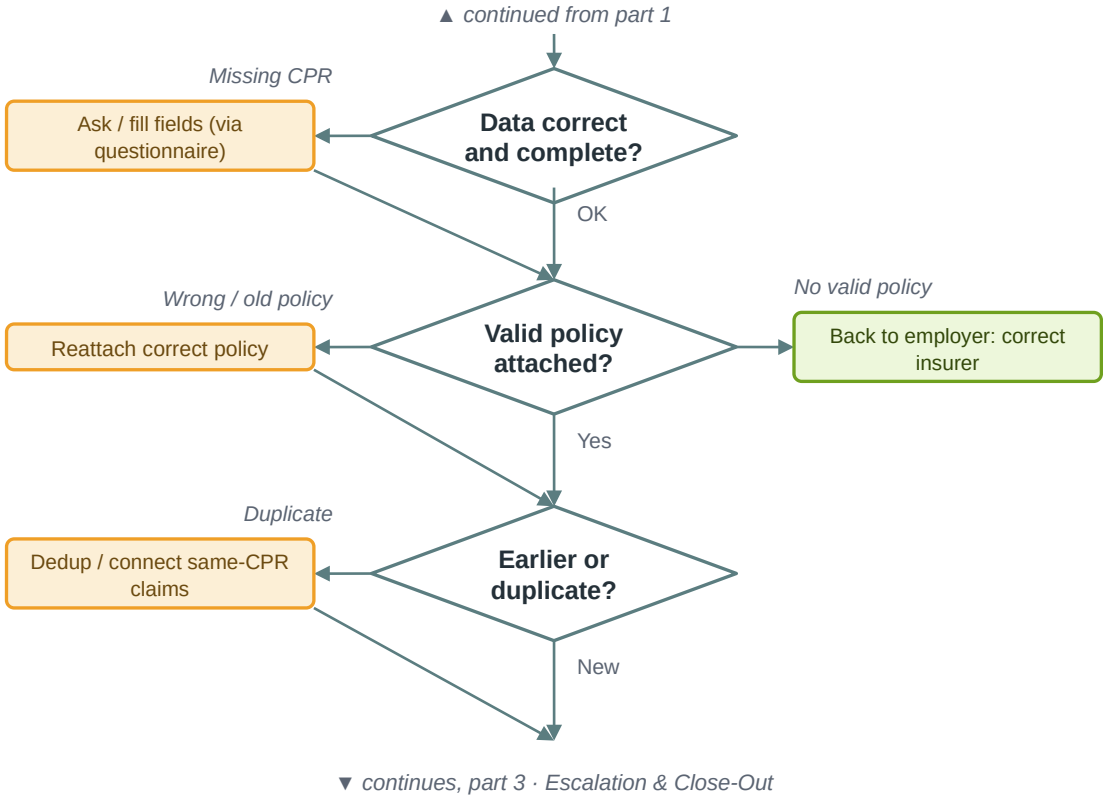
These two agents together replace 3 entry channels, the auto-import failure path, and the entire 1.1 side-track diagram.

Three checks done one after another become one check done all at once

Is it complete, linked to the right policy, and not already in the system? None of the three depends on the others

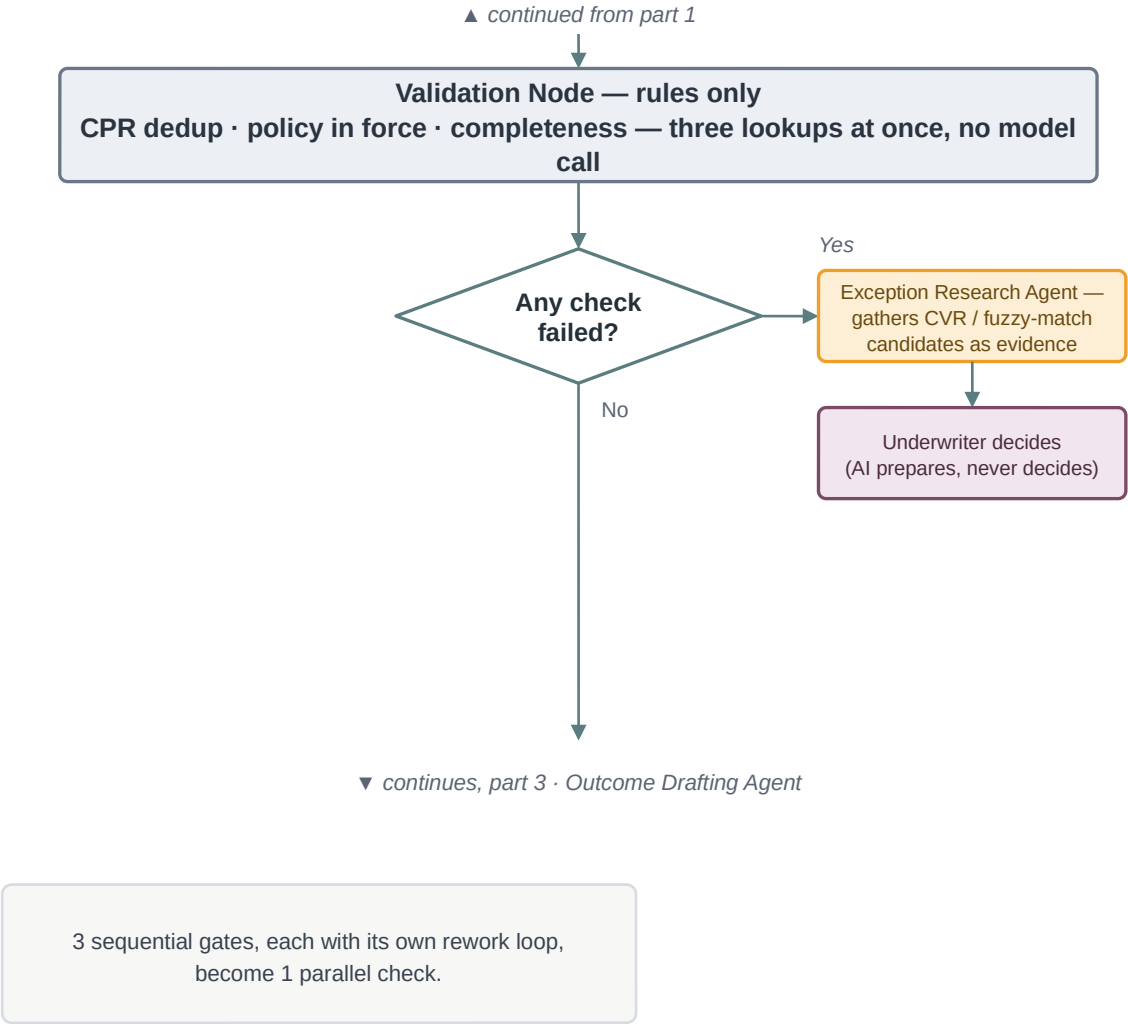
- Trigger / start
- Step / decision
- Exception / rework
- Exit

TODAY



- AI agent (LLM call)
- Rules only, no model
- Human checkpoint

AI-NATIVE — one LangGraph state graph · shared claim state carries the audit trail

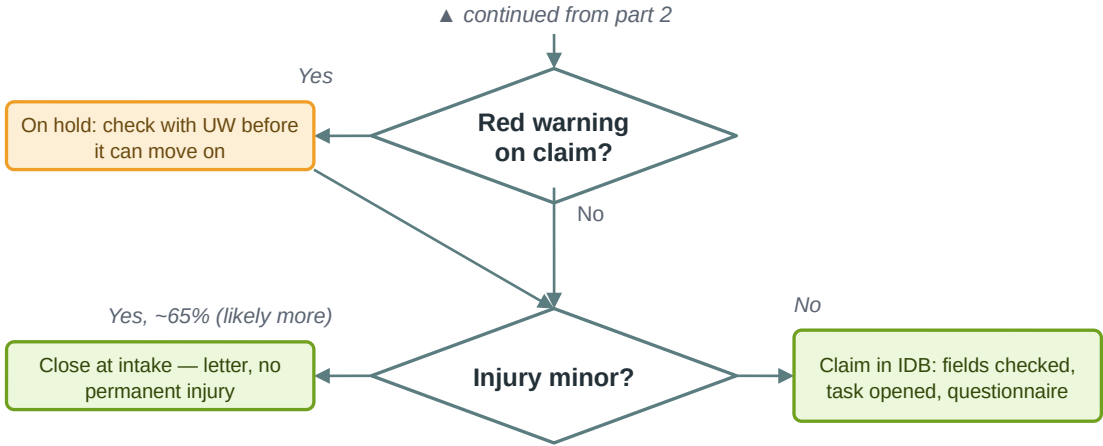


The AI writes the draft; the handler still makes the decision

From a system warning to a closed claim or a fully opened one — nothing that can go against a claimant is automated

- Trigger / start
- Step / decision
- Exception / rework
- Exit

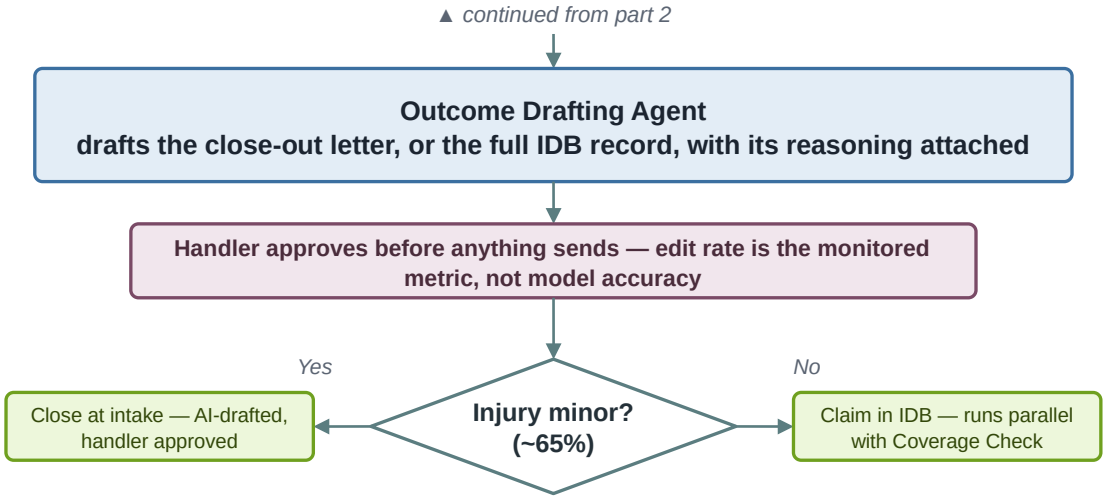
TODAY



Today: 7 decision gates, 3 entry channels, 10 exception or exit branches across the full FNOL flow.

- AI agent (LLM call)
- Rules only, no model
- Human checkpoint

AI-NATIVE — one LangGraph state graph · shared claim state carries the audit trail



Net result across all three parts:

4 LLM agents + 1 rules-only node · 2 human checkpoints kept by design · 3 entry channels → 1 · 3 sequential gates → 1 parallel check · shared claim state carries the audit trail

Coverage Check stops being a second investigation — the evidence is already in the claim

Every reason a claim can fail the coverage test, and where each one should be settled

Why coverage looks missing	Today	Redesigned	Who decides
Date sits inside a valid policy — no warning	quick manual check	rules, no model call	system
A valid follow-up policy exists	find it, reattach	registry lookup, reattach drafted	handler
Self-employed, voluntary cover	manual self-check	registry lookup, rules	handler
Group or branch policy not surfaced	ask underwriting	agent gathers the evidence, drafts the question	underwriting
Company-name mismatch	ask underwriting	agent fuzzy-matches CVR, drafts the question	underwriting
Wrong, expired, or no policy at all	back to employer	letter drafted, never sent unread	handler

Only two rows change hands. Four of the six are lookups against a register the insurer already queries — those become rules with no model call at all. The two that genuinely need judgement still go to underwriting, because that authority is not ours to take. The AI's job there is to arrive with the evidence assembled, not to answer the question.

WHAT THIS REMOVES

The second look at the policy	FNOL already asked “is a valid policy attached?”. Coverage Check asks it again, from the same documents.
The wait between the two	Today the claim sits in IDB until someone picks up the coverage step. It now runs beside claim creation, not after it.
Re-reading the file to ask underwriting	The question arrives with the policy history, the CVR match and the dates already attached.

Causes and their handling are taken from Diagram 2 of the brief - “who decides” is the design choice, not an observation

THE HANDOFF

The brief asks about the join between the two workflows. Today it is a queue. In the redesign there is nothing to hand over.

TODAY

- FNOL closes. The claim is written to IDB.
- It waits in a queue for the coverage step.
- A second person opens the same documents.
- They re-ask a question FNOL already asked.

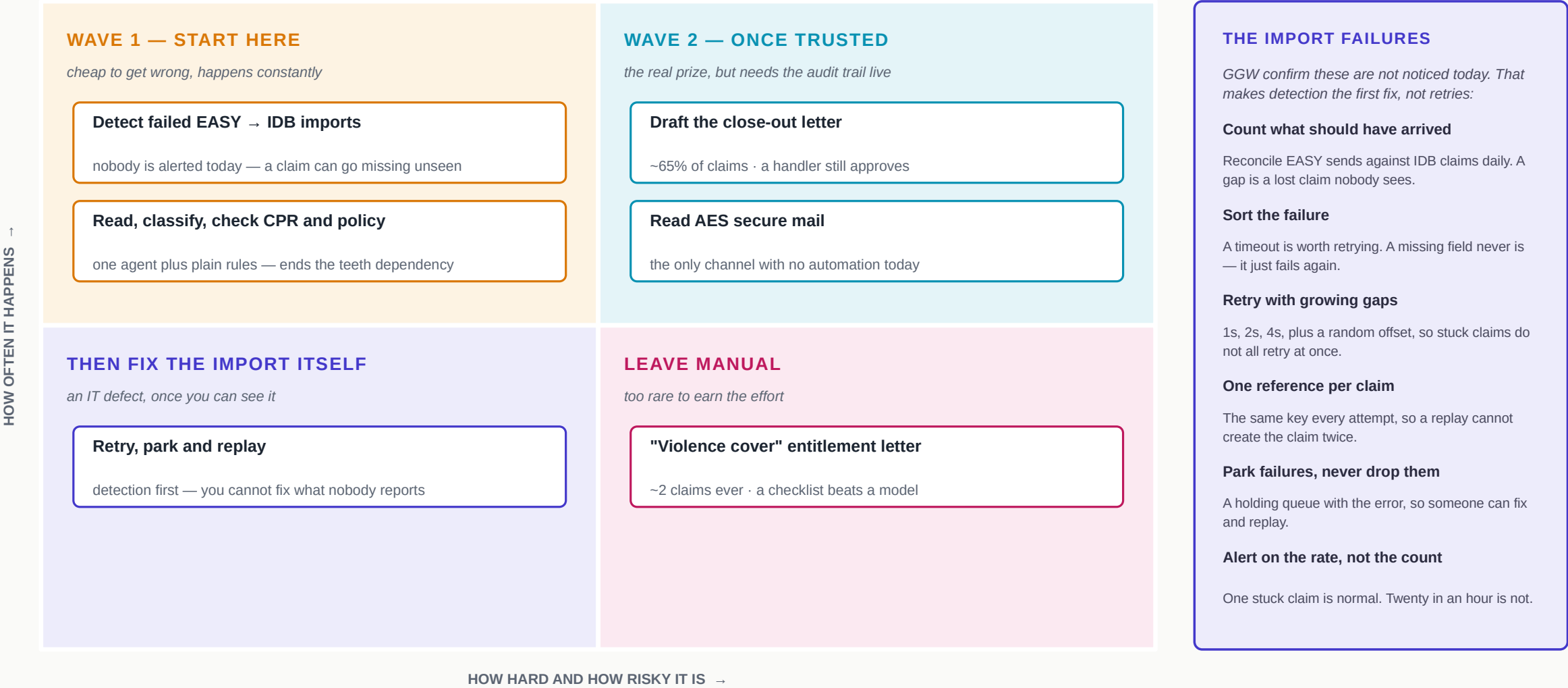
REDESIGNED

- One state graph spans both workflows.
- Extraction captures policy, dates, CVR and employment status once.
- Coverage is a rules pass over state that already exists.
- It runs beside claim creation, not after it.
- Only the exceptions become anyone's work.

A handoff you delete cannot be the thing that breaks.

Start where a mistake is cheap — that is what earns the trust the bigger wins depend on

“Hard” means build effort and regulatory exposure together, not just weeks of work



AES rulings and underwriting authority are deliberately absent — those decisions are not ours to make · frequencies from the case study

1 · SHADOW

wks 1–8

Agents run on live claims. Output is stored and compared with what the handler did. Nobody sees it.

Exit when extraction agrees with the handler on 95% of fields.

2 · SUGGEST

wks 9–16

The handler sees the draft and must edit or approve it. Nothing sends itself.

Exit when the edit rate on minor claims settles below 20%.

3 · APPROVE

mths 5–6

Minor claims arrive pre-filled. One click accepts. Everything else still drafted only.

Exit criteria reviewed monthly with the claims lead.

NEVER

—

No automatic decision on anything adverse to a claimant, on an AES ruling, or on underwriting authority.

This is a design boundary, not a phase.

WHAT WE WATCH — AND WHY THAT METRIC	
Edit rate on drafts	The trust signal. A model can be accurate and still not be used.
Field-level extraction precision	Checked against the source document, per field, not per claim.
Exception rate, by node	Says which step is actually struggling, not that “the AI” is.
Override rate by claim type	Feeds the confidence threshold. Teeth claims will differ from the rest.
First report to linked claim	The only number the business asked for. Everything else is diagnostic.
Alert on the rate, not the count — one stuck claim is normal, twenty in an hour is not.	

WHEN IT BREAKS

An agent fails, or confidence is low

The claim drops into today's manual queue. The old path is never removed, so the failure mode is a slower claim, never a wrong one.

A prompt or model version changes

200 historical claims are replayed first. A regression against the recorded outcome blocks the deploy.

One agent misbehaves in the wild

Each is behind its own switch. Turn off drafting and extraction keeps running.

The run dies half way

Graph state is checkpointed, so the claim resumes at the last good node instead of starting again.

Every decision, its inputs and its confidence are written to the audit log — that is what makes an adverse outcome defensible later.

Thresholds shown are the ones we would propose and then argue about with the claims lead — they are starting points, not findings

A · THE CLAIM STATE — WHAT EVERY NODE SHARES

```
class ClaimState(BaseModel):
    claim_id: str
    source: Literal["easy", "aes", "email"]
    documents: list[BlobRef]
    fields: ExtractedFields          # typed, per-field
    confidence: dict[str, float]     # per field
    claim_type: Literal["standard", "teeth",
                        "glasses", "violence"]
    coverage: CoverageResult | None
    exceptions: list[Exception]
    drafts: list[Draft]
    audit: list[NodeRun]             # who did what, when
```

Pydantic types are the contract, and confidence is held per field rather than per claim. A node that cannot produce a valid field returns null plus a reason — it never guesses to satisfy a schema.

B · THE STACK, AND WHY EACH PIECE

- LangGraph · Python**
Nodes are the agents, conditional edges are the decision diamonds on slides 2–4. The diagram is the code.
- Pydantic**
Typed state. Schema violations fail loudly at the node, not silently downstream.
- PostgreSQL**
LangGraph checkpointer plus the audit log. One store, so a claim's history and its state cannot drift apart.
- Azure Container Apps**
Two always-on for the API and worker, two scale-to-zero for burst. Cloud, as GGW deploy today.
- Azure Blob · Key Vault**
Claim documents and secrets. Documents never enter a prompt as raw bytes — only extracted text.
- Azure OpenAI**
Same model tier throughout, pinned by version. EU region keeps claim data in-region.

C · INTEGRATIONS — WHAT DECIDES THE TIMELINE

- EASY IN**
Reuse the existing auto-import. We add the reconciliation that tells you when it failed.
- AES secure mail IN**
Fully manual today. A mailbox reader is the single biggest unlock on the intake side.
- E-mail IN**
Same reader, different folder. Attachments to Blob, text to extraction.
- IDB OUT**
Write the claim, attach the policy, open the task. If there is no API, this is the long pole — not the AI.
- CPR / CVR registers BOTH**
Already queried by hand. Wrapped as tools the agents call, with results cached on the claim.

D · HOW IT SHIPS AND KEEPS RUNNING

Infrastructure as code	Bicep or Terraform. Nothing clicked into the portal, so the test environment is the production one minus data.
Azure DevOps pipelines	Already licensed. Lint, type-check, unit tests, then the replay set on every merge.
A flag per agent	Extraction, classification, research and drafting each toggle independently, per claim type.
Blue/green on the workers	A bad release drains rather than cuts. In-flight claims finish on the old version.
Model version pinned	An upgrade is a deliberate change that re-runs the replay set — never a silent one.

The models are the easy part. Whether IDB can be written to programmatically is the question that moves the date.

Runs on the cloud GGW already deploy to · costs for exactly this shape are on slide 9 · the rollout that puts it in front of a handler is on slide 7

A · INTAKE TIME — MINUTES PER CLAIM, PER PERSON

Step	Today	Redesigned
Read the report, re-read it, open attachments	10.0	0.7
Take notes, check fields, chase a missing CPR	6.1	2.8
Attach policy · duplicate check · coverage date	5.0	0.0
Decide, draft and send / open the full claim	5.5	3.0
Work an escalation or an underwriting hold	1.0	0.6
Switch between EASY, IDB, e-mail and paper	3.2	0.7
Touch time	30.8	7.8
Breaks, fatigue, interruptions, rework (+30%)	9.2	2.3
Standard time — what the design allows	40.0	10.1
What the business case actually banks	40.0	19.0

The design allows 10.1 min a claim. We do not bank that. The case uses a three-point estimate — (best 30 + 4 × likely 21 + worst 12) ÷ 6 = 21.0 min saved a claim — which is the 19.0 min above, and the 19 minutes on slide 1. In plain terms: we assume handlers capture 70% of the saving the design allows.

Cross-check: 40 min × 12,500 claims a year = 8,332 hours — 52% of the 10-person support team's year, 17% of all 30. Per person: 5 claims a day at 40 min is 3.3 of a 6.4-hour day. Intake is half of what the support team does.

D · THE BUSINESS CASE

EVERY YEAR

Capacity released	4,370 support-h · 2.8 FTE
Worth, at €31.83 per support-team hour	€139,100 a year
Cost to run it — AI models and Azure	– €2,400
Ongoing support — 0.25 of an engineer	– €24,400
Net value, every year	€112,200

ONE-OFF

Build — 3 people × 6 months, fully loaded	€146,400
Payback	16 months

The build is loaded team capacity, not new cash — the AI team is already funded. New cash needed: €2,400 a year. A shorter build costs less: 4 months ≈ €98k.

B · AI COST TO PROCESS ONE CLAIM

Agent	Tokens in	Out	Runs	US\$ / claim
Extraction	11,500	700	100%	0.0314
Classification	900	120	100%	0.0032
Validation — rules only, no model	—	—	100%	0.0000
Exception research	2,500	400	20%	0.0020
Outcome drafting	2,200	550	100%	0.0110
Subtotal				0.0476
× 1.75 for retries, guardrails, evals				0.0833
Per claim				≈ €0.077

One claim ≈ a scanned form, an e-mail and 3–4 attachments. All four agents run on one model tier — gpt-5.6 terra, \$2 in / \$12 out per million tokens.

C · AZURE, WEST EUROPE — MONTHLY

Component	US\$/mth
4 container apps — 2 always on, 2 burst only	55
PostgreSQL B2s + 64 GiB storage and backup	61
Blob storage for documents + audit logs	8
Net new, incl. ~8% West Europe premium	134

Registry, networking, Key Vault and DevOps (≈ €670/yr) already exist, so they are not charged here. Azure DevOps gives repos and pipelines, not a database — PostgreSQL is genuinely new.

E · WHO GETS THE TIME BACK — CONFIRMED BY GGW

20 claim handlers and 10 support staff (GGW, 8 Sep). The brief puts FNOL with the support team, so every freed hour is theirs.

Per support-team member	Today	Redesigned	Difference
Claims taken in, per day	5	5	—
Time on intake, per day	3.3 h	1.6 h	– 1.75 h
Share of a productive day	52%	25%	– 27 pts
Hours back, each, per day			1.75 h

Who does intake

20 claim handlers, 10 support staff — FNOL sits with support

GGW (P. Gossmann), 8 Sep 2026

Failed EASY imports

Not noticed today — so detection comes before retries

GGW (P. Gossmann), 8 Sep 2026

Volumes and shares

~50 first reports a day · 65% minor · ~30 in the team

GGW case study brief

Wage base — intake

Sachbearbeiter/in Versicherung, €41,100 gross

StepStone Gehaltsreport

Employer on-cost

≈ €23 per €100 gross — health, pension, care

Lohnnebenkosten 2026, sevdesk

Working time, breaks

38 h tariff week; statutory break rules

Arbeitszeitgesetz §4

Build & upkeep pay

Senior ML engineer ≈ €98k; juniors at €70k as briefed

Glassdoor, ML Engineer Germany

Model prices

gpt-5.6 terra — \$2 in / \$12 out per million tokens

OpenAI API pricing

Container hosting

Consumption plan rates and the monthly free grant

Azure Container Apps pricing

Database

PostgreSQL Flexible Server B2s — 2 vCore / 4 GiB

Azure Database for PostgreSQL

Upkeep after go-live

15–20% of build effort a year — our 0.25 FTE is 17%

Software maintenance rule of thumb

Industry baseline

17 of 20 carriers still manual; none let AI decide alone

Decerto, 2026 Claims Decisioning Pulse

Where value comes from

The 10-20-70 split — algorithms, technology, people

BCG, AI in customer service operations

Fixing import failures

Retry vs. dead-letter, idempotency keys, alert on rate

Dead-letter queues and poison messages

STILL OUR ASSUMPTIONS — AND THE FIRST THING WE WOULD TEST

Every per-step minute on slide 9 · the 30% allowance for breaks and interruptions · how often a CPR is missing (30%) or underwriting is asked (25%) · how often the redesign raises an exception (20%) · the tokens a typical claim needs · the €146k build · the 0.25 FTE upkeep.

Who does intake is no longer among them — GGW answered it, and every per-person figure in this deck is rebuilt on that answer. Two weeks of watching live intake would replace the rest with measurements.

confirmed by GGW or stated in the brief

published source, linked

Prices checked September 2026 · US\$ converted at €0.92 · every euro figure is fully loaded — gross pay plus employer contributions